

Exact Posterior Probability of Optimality for Factor-Gaussian Beliefs

The Integral That Thompson-Style Methods Approximate,
Computed in Closed Form at Scale

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Abstract

The posterior probability that each alternative is best drives stopping rules, budget allocation and Thompson sampling, and is widely treated as intractable. Under common random numbers, and under per-arm Gaussian observations with a factor prior, the posterior covariance is exactly a low-rank factor term plus a diagonal at every step. For such a covariance the whole probability vector is computable exactly, for all n alternatives at once, in time linear in n . In a ranking-and-selection experiment, stopping rules built on marginal means and variances spend up to three and a half times as many replicates as the exact rule to reach the same error. In a correlated bandit, sampling from the exact vector matches Thompson sampling, while a variational policy built on marginals pays more regret.

1 Introduction

Let $\theta \in \mathbb{R}^n$ be the unknown values of n alternatives, such as simulated systems, bandit arms or candidate molecules, and let $\mathcal{D}$ be the data. The posterior probability of optimality is the vector

$$p_i = \Pr\left(\theta_i = \max_j \theta_j \mid \mathcal{D}\right), \quad i = 1, \dots, n. \quad (1)$$

Sequential decision rules use this vector directly. A stopping rule halts when $\max_i p_i$ reaches a confidence target. A budget allocator spends where several p_i are comparable. Thompson sampling plays an action with probability p_i without computing it. A policy with constraints, or one that must be derandomized, needs the vector itself, because one posterior sample does not determine it.

The Bayesian reinforcement learning literature describes Equation (1) as intractable. [Tarbouriech et al. \(2023\)](#) call it the key object of the field and write that computing it “involves computing several complicated integrals with respect to the posterior and is intractable in most cases.” Their method, VAPOR, replaces it with a variational program over occupancy measures.

[Menet et al. \(2026\)](#) fine-tune a large language model toward the same object under a Gaussian-process posterior. The tractable policy they target uses only the marginal means and marginal standard deviations of that posterior. They note that the guarantee available for this family assumes “the worst-case bandit with independent arms.”

In ranking and selection, procedures bound the probability of correct selection with Bonferroni-type inequalities or estimate it by posterior sampling. The joint vector is not evaluated.

In several of these settings the posterior covariance has the exact form

$$\Sigma = WW' + \text{diag}(d), \quad W \in \mathbb{R}^{n \times \rho}, \quad \rho \ll n. \quad (2)$$

This form is not a modelling approximation. It follows from how the data are generated. For a covariance of this form, Equation (1) is computable exactly for every i at once, in time linear in n .

The computation is the pricing pass for correlated contests in Cotton (2026), which also develops its derivatives and its inverse and benchmarks it to $n = 10^6$. This paper contributes three things. Section 3 shows that two standard posteriors keep the form of Equation (2) after every update. Section 4 identifies the approximations in current use and the identity that ties the exact vector to Thompson sampling. Sections 5 and 6 measure what the exact vector changes in a stopping problem and in a bandit.

2 Computing the vector for a factor-form covariance

Write $\theta = m + Wz + \varepsilon$, with $z \sim N(0, I_\rho)$ and independent $\varepsilon_i \sim N(0, d_i)$. Given z , the alternatives are independent, with means $m_i + w_i'z$ and variances d_i . Let $F_{i|z}$ and $f_{i|z}$ be the conditional distribution and density functions. Then

$$p_i = \int_{\mathbb{R}^\rho} \phi_\rho(z) \int_{\mathbb{R}} f_{i|z}(x) \prod_{j \neq i} (1 - F_{j|z}(x)) dx dz. \quad (3)$$

The inner integrands for different i share the product $G_z(x) = \prod_j (1 - F_{j|z}(x))$. Dividing out the i th term gives every p_i from the same product:

$$p_i = \mathbb{E}_z \int f_{i|z}(x) \frac{G_z(x)}{1 - F_{i|z}(x)} dx. \quad (4)$$

On a lattice of L points in x , with Q quadrature nodes in z , the whole vector costs $O(nLQ)$.

The numerical details are in Cotton (2026): lattice placement, products taken in logs, and the tails. That implementation is used here without change. Every exact vector reported below was checked against argmax frequencies from 2×10^5 Monte Carlo draws. The largest total-variation distance in those checks is 0.0042, which is at the Monte Carlo noise level for that sample size.

3 Two posteriors with exact factor form

3.1 Common random numbers

Ranking and selection compares n simulated systems. Under common random numbers, replicate r runs every system on the same random scenario:

$$Y_{ir} = \theta_i + v_i' F_r + \varepsilon_{ir}, \quad F_r \sim N(0, I_\rho), \quad \varepsilon_{ir} \sim N(0, \sigma_i^2). \quad (5)$$

The scenario effect F_r is the same for every system. The loadings v_i and noise scales σ_i are known properties of the simulation. The unknown is θ , with prior $\theta \sim N(0, s_0^2 I)$.

The covariance of one replicate across systems is

$$\Lambda = VV' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_n^2). \quad (6)$$

After n_r replicates with mean vector $\bar{Y}$, the posterior of θ is Gaussian with precision

$$A = s_0^{-2}I + n_r\Lambda^{-1}. \quad (7)$$

The Woodbury identity gives

$$\Lambda^{-1} = D^{-1} - D^{-1}VM^{-1}V'D^{-1}, \quad M = I_\rho + V'D^{-1}V. \quad (8)$$

With $U = D^{-1}V$ and $S = n_rM^{-1}$, the precision becomes

$$A = \text{diag}(a) - USU', \quad a_i = s_0^{-2} + n_r\sigma_i^{-2}. \quad (9)$$

This is a positive diagonal minus a positive semidefinite term of rank ρ .

Applying the Woodbury identity again, to Equation (9), gives

$$A^{-1} = \text{diag}(a)^{-1} + \text{diag}(a)^{-1}UCU'\text{diag}(a)^{-1}, \quad C = (S^{-1} - U'\text{diag}(a)^{-1}U)^{-1}. \quad (10)$$

The matrix C is positive definite. To see this, let $K = \begin{pmatrix} \text{diag}(a) & U \\ U' & S^{-1} \end{pmatrix}$. The Schur complement of S^{-1} in K is $\text{diag}(a) - USU' = A$, which is positive definite, and S^{-1} is positive definite, so K is positive definite. The Schur complement of $\text{diag}(a)$ in K is C^{-1} , so C^{-1} is positive definite too.

The posterior covariance is therefore

$$\Sigma_{\text{post}} = \text{diag}(d) + WW', \quad d_i = a_i^{-1}, \quad W = \text{diag}(a)^{-1}UC^{1/2}, \quad (11)$$

exactly, for every n_r . The rank of W is ρ , the number of shared scenario streams, not the number of systems. The posterior mean is $m = A^{-1}h$ with $h = n_r\Lambda^{-1}\bar{Y}$, and both factors are available from Equations (8) and (10). No step approximates. The posterior has factor form because the scenario noise has factor form.

3.2 Per-arm observations with a factor prior

In a finite Gaussian bandit the correlation sits in the prior on the arm means: $\theta \sim N(0, \Sigma_0)$ with $\Sigma_0 = VV' + \text{diag}(d_0)$. Linear models $\theta_i = v_i'\beta + e_i$ produce this form, and so do inducing-point Gaussian-process posteriors and the heteroscedastic output layers of large classifiers. Pulling arm a returns $\theta_a + N(0, \sigma_n^2)$.

With per-arm pull counts c , the posterior precision is

$$A = \Sigma_0^{-1} + \sigma_n^{-2} \text{diag}(c). \quad (12)$$

Equation (8) applied to Σ_0 puts A in the form of Equation (9), with $a_i = d_{0i}^{-1} + c_i\sigma_n^{-2}$, $S = M_0^{-1}$ and $M_0 = I_\rho + V'\text{diag}(d_0)^{-1}V$. The rest of the argument is unchanged. The posterior is exactly factor plus diagonal after every pull, with the same rank ρ as the prior.

4 Approximations in current use, and Thompson sampling

Two computable rules currently stand in for Equation (1). Both use only the marginal means m_i and marginal standard deviations $s_i = \sqrt{d_i + \|w_i\|^2}$.

The *independence approximation* treats the posterior as $\prod_i N(m_i, s_i^2)$. Menet et al. (2025) compute its probabilities of being best with a threshold κ^* chosen so that $\sum_i \Phi((m_i - \kappa^*)/s_i) = 1$.

The *VBOS policy* is the variational Bayesian optimistic sampling rule of O’Donoghue and Lattimore (2021), extended by Tarbouriech et al. (2023). In the near-closed form of Menet et al. (2026) it is

$$\tilde{\pi}_i = v\left(\frac{m_i - \kappa^*}{s_i}\right), \quad v(c) = \exp\left(-\frac{1}{8}(\sqrt{c^2 + 4} - c)^2\right), \quad (13)$$

with κ^* again chosen so the vector sums to one.

The VBOS policy departs from p on purpose, and its regret analysis depends on that departure. It is a policy, not an estimate of p . Its distance from p is reported below because the fine-tuning literature uses it as the trainable substitute for p .

Thompson sampling draws $\tilde{\theta}$ from the posterior and plays the argmax of the draw. Its probability of playing arm i at any history is therefore p_i at that history. Lemma 8 of Tarbouriech et al. (2023) states the general version: the expected occupancy of Thompson sampling equals the posterior probability of optimality.

Sampling an arm from the exact vector, called *exact matching* below, is therefore Thompson sampling in distribution. The two must have the same expected regret, and Section 6 checks that they do. The exact vector adds a computed object, which a stopping rule, an allocator or a derandomized policy can use where a single draw cannot.

5 Stopping under common random numbers

The experiment uses the posterior of Section 3.1 with $n = 100$ systems, $\rho = 2$ scenario streams and prior scale $s_0 = 1$. True means are drawn from the prior. Every system has replicate variance one, so each replicate carries the same marginal information in every regime. The *factor share* is the fraction of that variance that comes from the shared scenario.

The regimes differ only in how systems load on the scenario streams:

- *Independent*: factor share zero. This is the control.
- *Aligned*: every system loads on the same stream, with factor share 0.6. Differences between systems then vary less than their marginal variances suggest.
- *Opposed*: half the systems load positively and half negatively on one stream, with factor share 0.6. Differences between systems in opposite halves then vary more than their marginal variances suggest.

Three rules compete. Each computes its own estimate of p after every replicate, stops the first time its largest entry reaches $1 - \delta$ with $\delta = 0.05$, and selects that entry. The exact rule uses the computation of Section 2. The other two use the independence approximation and the VBOS policy.

Table 1 shows that every rule held its nominal error. The exact rule’s 0.944 in the opposed regime is within binomial noise of 0.95, with 179 stopped runs and a standard error of 0.016.

The two marginal-only rules held the error by spending more. They used 1.5 to 3.4 times the replicates of the exact rule and hit the budget far more often. The gap is largest in the aligned regime. There the leading systems’ differences are tightly coupled, and the exact rule stops after 69.7 replicates on average, against 175.6 and 240.1.

The marginal-only estimates are far from the exact vector along the way. Before stopping in the aligned regime, the independence approximation is at median total-variation distance

regime	rule	mean replicates	realized PCS	censored of 200
independent	exact	141.2	0.970	35
	independence	207.0	0.985	65
	VBOS	271.5	1.000	102
aligned	exact	69.7	0.958	9
	independence	175.6	1.000	60
	VBOS	240.1	1.000	91
opposed	exact	106.3	0.944	21
	independence	171.7	0.979	57
	VBOS	239.0	0.991	88

Table 1: Stopping at nominal 95% confidence. Each regime has 200 replications and a budget of 400 replicates. A run that reaches the budget without stopping is censored and counts 400 toward the mean. Realized PCS is the fraction of stopped runs that selected the true best system.

0.123 from the exact vector, with ninetieth percentile 0.174. The VBOS policy is at 0.141, with ninetieth percentile 0.206.

The distances do not vanish in the independent regime, where they are 0.063 and 0.096. The independence approximation uses a shared threshold and is biased even when the posterior is independent. The VBOS policy departs from p by design. The exact rule’s estimate matched Monte Carlo to within total-variation distance 0.0015 throughout.

6 A correlated Gaussian bandit

The bandit uses the posterior of Section 3.2 with $n = 50$ arms, prior variance one per arm, observation noise $\sigma_n = 1$, horizon 200 and 100 replications. The prior has rank two. In the *independent* regime it has no factor term. In the *clusters* regime each half of the arms loads on its own factor, which carries 0.75 of the prior variance.

Four policies act on the same posterior: Thompson sampling, exact matching, sampling from the VBOS policy, and sampling from the independence approximation. Table 2 reports cumulative Bayes regret and, at the horizon, whether the arm with the highest posterior mean is the true best and the shortfall of that arm’s value below the best. Its last column is the total-variation distance between each policy’s action distribution and the exact vector at mid-horizon, along that policy’s own trajectory.

Thompson sampling and exact matching differ by one to two percent of cumulative regret in both regimes, within replication noise, as Lemma 8 requires. The VBOS policy pays ten to eleven percent more regret than Thompson sampling, and its action distribution is at total-variation distance 0.13 to 0.16 from the exact vector. The independence approximation pays four to five percent more.

The identification rates do not separate the policies. Their differences are within two binomial standard errors at 100 replications. The regret and distance columns carry the comparison.

regime	policy	regret	identification	shortfall	TV to exact p
independent	Thompson	172.1	0.72	0.073	0
	exact matching	173.9	0.75	0.055	0
	VBOS	190.3	0.76	0.061	0.159
	independence	179.2	0.79	0.052	0.057
clusters	Thompson	122.7	0.61	0.097	0
	exact matching	124.8	0.51	0.113	0
	VBOS	135.9	0.59	0.105	0.129
	independence	128.7	0.51	0.121	0.061

Table 2: Mean over 100 replications of cumulative Bayes regret, the rate at which the highest posterior mean identifies the true best arm at the horizon, and the shortfall of that arm. The last column is the total-variation distance from the exact vector at mid-horizon. It is zero for Thompson sampling by Lemma 8 and for exact matching by construction.

7 Scope

The derivations of Section 3 are exact because the observations there are generated with factor structure. When scenario effects enter a simulation nonlinearly, a low-rank model of Λ is an approximation, and the posterior inherits its error. When a Gaussian-process kernel is dense with a short length scale, a factor fit has to precede the computation, and its residual is the cost.

Under simple-regret loss, the one-step value of a measurement is the knowledge-gradient quantity (Frazier et al., 2008). Tuisov (2026) shows that this is the normative myopic value of computation, and notes that the exact dynamic version “is itself generally intractable.” The method here does not compute that dynamic value. For factor-Gaussian beliefs it makes the myopic quantities and the stopping certificates exact.

The regret analysis of O’Donoghue and Lattimore (2021) depends on the VBOS tilt. Its measured distance from p quantifies the approximation, but does not by itself make a case against the tilt.

Thompson sampling needs one posterior sample and no vector. The exact vector matters where a sample is not enough: stopping, allocation, constraints and derandomization. Both experiments above are of that kind.

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